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# RESIDUAL CLAIMS AND INCENTIVES IN RESTAURANT CHAINS

**by**

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**CES 06-18** **July, 2006**

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Abstract

I examine the relationship between ownership and production activities using a new dataset of restaurant chains. Production in restaurant chains provides an opportunity to examine the effects of residual claims on incentives because production is decentralized and fairly uniform across restaurants in the same chain. Yet the allocation of residual claims varies between company-owned and franchised units, affecting the strength of incentives for restaurant-level activities. The decision to own or franchise each restaurant reflects the value of either withholding or allocating residual claims for performing these activities. I find that more complex production activities are systematically correlated with company ownership. Onsite food production raises the likelihood of company ownership by 28% relative to offsite food production. Table service raises the likelihood of company ownership by 26% relative to counter service. The results are not consistent with straightforward effort-promoting effects of residual claims in simple principal agent models. They are consistent with the view that residual claims can generate unbalanced incentives across diverse tasks.

* I would like to thank Thomas Hubbard, Ali Hortacsu, Chad Syverson, Francine Lafontaine, Raphael Thomadsen, Thomas Holmes and Robert King for helpful comments and discussion, as well as seminar participants at the University of Chicago, IIOC 2005 and University of Minnesota for their input. I would also like to thank Lynn Riggs for support and assistance at the Census Bureau. Financial support from the Esther and T.W. Schultz Endowment Fund and Francis William Immasche Endowment Fund at the University of Chicago are gratefully acknowledged.

*Disclaimer:* The research in this paper was conducted while the author was a Census Bureau research associate at the Chicago Census Research Data Center. Research results and conclusions expressed are those of the author and do not necessarily indicate concurrence by the Bureau of the Census. This paper has been screened to insure that no confidential data are revealed. Support for this research at the Chicago RDC from NSF (awards no. SES-0004335 and ITR-0427889) is also gratefully acknowledged.

# 1 Introduction

A central question in economics is whether production should be organized within a firm or by market transactions. Coase (1937) observed that production within a firm can be more efficient than market transactions. Recent theories such as Holmstrom and Milgrom (1991, 1994) and Hart, Shleifer and Vishny (1997) show that transactions within a firm among agents with few residual claims can achieve greater efficiency by balancing incentives for different types of activities where market forces may distort them. Recent empirical work by Shepard (1993), Slade (1996), Hubbard (2000), Baker and Hubbard (2003, 2004), Lafontaine and Shaw (2005) shows that ownership decisions in various industries reflect the value of muting market incentives when it is difficult to control specific activities. This paper finds evidence that such concerns affect the distribution of residual claims between chain companies and franchisees in the restaurant industry. An empirical analysis of establishment-level ownership and restaurant characteristics reveals that restaurants with more production activities are more likely to be company-owned, implying that managers with more responsibilities generally have fewer residual claims. This is not consistent with the view that franchising solves simple moral hazard problems, which would predict the opposite, namely, that managers with more responsibilities should be motivated with stronger incentives. Instead, the results suggest that it may be efficient to limit the strength of managers' incentives when it is more difficult to control their decisions.

Production in restaurant chains provides an opportunity to examine the effects of residual claims on incentives because production is decentralized and fairly uniform

across restaurants in the same chain. Yet some restaurants are company-owned and some are franchised, generating different incentives for similar activities. The main difference between company-ownership and franchising is that the employee-managers of company-owned units have few residual claims and weak incentives to maximize restaurant profits while franchisees retain a large share of profits and therefore, have strong incentives to maximize profits at their restaurants. The goal of this paper is to directly compare the choice between ownership and franchising to the types of activities carried out at each restaurant to determine if weak or strong market incentives are efficient for motivating different types of restaurant production.

The analysis exploits a new establishment-level dataset from the 1997 Census of Retail Trade that contains comprehensive information about the population of restaurants in scores of the largest national chains and detailed information on the characteristics of individual restaurants. I compare characteristics of the production process at individual chain restaurants to their organizational form. Two attributes that vary across chains are the extent of onsite food preparation and type of customer service. Characteristics that vary within chains and service categories are the takeout, drive-through and delivery service, the availability of seating and average meal price. I find that onsite food production raises the likelihood of company ownership by 28% relative to offsite food production and table service raises the likelihood of company ownership by 26% relative to counter service. I also find that limited service restaurants that primarily generate takeout or drive-through and delivery service are, respectively, 13% and 22% less likely to be company-owned than restaurants with dine-in services. Limited service restaurants

that provide seating are 18% more likely to be company-owned than those with no seating. Full service restaurants with average prices over $10 are 24% more likely to be company-owned than those with prices below $10. These results indicate that weak incentives associated with company ownership are systematically correlated with more complex production activities. I find little evidence for non-incentives-based explanations such as regulatory and risk differences across markets, idiosyncratic chain effects, chain size and vintage measures that reflect factors such as credit constraints and brand value. This suggests that weak incentives are optimal for more complex production activities, perhaps because it is more difficult to balance managers' efforts across different activities.

The literature on agency problems and contracting difficulties provides a framework for interpreting my results. Residual claims play a role in motivating workers when it is impossible to link rewards directly to effort. In simple principal-agent models such as Caves and Murphy (1976), Rubin (1978) and Matthewson and Winter (1985), workers' efforts are not observable, causing them to shirk unless they share a stake in profits. Residual claims can generate effective incentives for workers but force them to bear more risk or generate moral hazard on the part of the principal. In more complex principal-agent models such as Holmstrom and Milgrom (1991, 1994) and Hart, Shleifer and Vishny (1997), workers have disparate tasks and the structure of residual claims matters because it is difficult to adequately link rewards to different actions. In these models, performance-based incentives serve not only to motivate hard work, but also to allocate a worker's efforts among his responsibilities. Granting a worker residual claims may skew

his efforts in favor of activities that are more readily observable or directly rewarded by the market. Allocating his effort in this way raises his private gain but may lower overall efficiency. This is a broad feature of models in which agents have imperfectly-aligned incentives and discretion in allocating their efforts.

These features are relevant to chain restaurant production since franchisees may have unique incentives and it is probably impossible to specify and verify activities that take place at individual restaurants in all possible circumstances. The findings in this paper indicate that managers with more responsibilities generally have fewer residual claims, suggesting that company ownership may be a means to suppress incentives for activities that cannot be directly discouraged if the scope for such activities increases with production complexity. The results do not support the idea that strong incentives via franchising motivate only efficient effort in chain restaurants. Other studies such as Lafontaine (1992), Kalnins and Lafontaine (2001) and Lafontaine and Shaw (2005) recognize and examine incentives-related costs to franchising such as moral hazard on the part of franchisors, free-riding incentives and spillover effects within chains. The results in this paper further support the view that distorted incentives may be a drawback to franchising. Company ownership may serve to neutralize the negative effects of strong incentives when restaurant production is more difficult to monitor and control.

The remainder of the paper is organized in the following way. The next section provides a conceptual framework for comparing the types of tasks and incentives of managers in chain restaurants. Section 3 describes data sources and variables in the analysis. Section 4 lays out the empirical framework and describes results from cross-

chain and within-chain variation in production technology. Section 5 provides concluding remarks. The appendix contains tables of results.

## 2 Conceptual Framework

This paper compares the range of production activities in restaurants to the choice of organizational form for these activities in order to draw inferences about how residual claims affect incentives for restaurant managers. In Subsection 2.1, I describe the types of production activities that typically take place at restaurants, emphasizing how different activities add to or reduce a restaurant manager's responsibilities. Subsection 2.2 describes managers' roles and compensation under company ownership and franchising, pointing out that the main difference lies in residual claims. Subsection 2.3 draws from the theoretical literature to provide a framework for interpreting the link between production activities and residual claims incentives. Principal-agent models of moral hazard examine the tradeoff between strong incentives that motivate hard work and strong incentives that lead to misallocation of effort towards private gains.

### 2.1 Production and Responsibilities in Restaurant Chains

Individual outlets within a chain of restaurants carry out similar production activities to make a standardized product. The amount of food preparation and mode of customer service at each restaurant in a chain depends on the business concept chosen by the chain

company. Full service chains customize food preparation for each order. Fast-food chains carry out standardized and often, automated food preparation. Ice-cream and donut shops carry out virtually no food preparation if they sell food products made at other locations. In terms of production activities, full service chains have the most complex food preparation activities, followed by fast-food chains, with simple snack shops such as ice-cream and donut vendors undertaking the least complex food preparation tasks.

The mode of customer service also varies across different types of chains. Full service restaurants provide table service; waiters offer personalized service to each customer. Fast food outlets and snack shops provide counter service; employees take orders and transfer food items to customers but provide little other direct customer service. Full service restaurants with table service typically undertake more complex customer service tasks than limited service restaurants such as fast-food outlets and snack shops.

While type of food service is the same for all restaurants in a chain, production modes may vary across restaurants in the same chain. For limited service chains, some outlets primarily serve takeout, drive-through or delivery customers, while others mainly serve dine-in customers. I shall assume that production with dine-in service generates more complex tasks than production for off-site consumption. With dine-in, there is more customer service and restaurant managers may have more scope to affect the customer's dining experience. With off-site consumption, customers probably value speed and consistency more than subjective features like ambience that depend on individual

managers' input and are more difficult to specify. In a similar vein, some restaurants in a limited service chain may offer no seating while others in the same chain provide seats and tables for customers. I shall assume that a lack of seating indicates more off-site than on-site consumption, which in turn indicates less complex production activities. The chain company typically controls the choice of production mode at individual restaurants. Franchise contracts generally limit franchisees' ability to freely modify restaurant services, especially of easily observable features such as the addition or removal of seating. In practice, however, the extent to which different services are produced may depend on joint decision making with franchisees.

The scope of production activities also varies widely across full service chains. Higher quality chains often undertake more complex food preparation tasks and more intensive customer service and also charge higher prices for each meal. I shall assume that higher-priced full service restaurants have more complex production activities than lower-priced restaurants. While franchisees have limited scope for changing the business format, they are legally entitled to set prices. In company-owned restaurants, the chain company sets both the business format and prices. Typically, there is some variation in prices within chains due to market conditions and possibly, also due to different strategic incentives. I focus on price variation across chains to examine the effect of quality variation across chains.

Each restaurant is supervised by a manager who must organize and coordinate all aspects of restaurant production. The manager's responsibilities grow in scale and scope as food preparation and customer service functions are added to a restaurant. Food

preparation generates tasks such as organization of food supplies, maintenance of cooking equipment, management of kitchen staff, adherence to health code regulations and coordination with customer service. Customer service requires tasks such as the management of counter or wait staff, maintenance of dining facilities and equipment, coordinating with the kitchen and generating goodwill with customers in general. Table service increases the importance and scope of customer service tasks. Takeout, drive-through and delivery-only outlets involve less customer service, as do restaurants with no seating facilities. Full service restaurants with higher prices tend to engage in more complex food production and more customer service.

## 2.2 Residual Claims in Company-Ownership and Franchising

One way by which chains can control managers' incentives is to withhold or grant residual claims in the form of employment or franchise contracts. A typical employment contract for restaurant managers consists of fixed salary payments and few performance bonuses. Brickley and Dark (1987) and Krueger (1991) find that a lack of incentive pay for company managers is common in the restaurant industry and other industries with franchising. Promotion within the company may provide performance-based incentives for employees but these will be aligned to chain objectives, unlike residual claims for franchisees which are outlet-specific. Brown's (1998) model of franchising also shows that promotion-based incentives are weak if there are relatively many low level managers and few high level positions to aspire to, as is the case in large national restaurant chains.

Moreover, this is a poor way to reward performance if the skills required of low and high level positions are different, as is likely for restaurant and corporate managers.

A franchisee usually takes an active role in the day to day operations of his restaurant and retains residual profits after paying a royalty rate of approximately 10% on revenues to the chain company. He has strong market-driven incentives for managing restaurant operations. A typical franchise contract gives a franchisee the right to operate a standardized restaurant at a particular location, with specific technology and proprietary trademarks, in exchange for a one-time franchise fee and monthly royalty payments. It typically requires a commitment from the franchisee for direct involvement in the operations of his restaurant through "on-premises" clauses or by limiting the number of partners involved in the franchise contract. It may also contain terms for the procurement of inputs and equipment, audit procedures, payments to central advertising funds, assistance and training programs, rights of transfer, renegotiation and termination clauses. The franchisee is free to set prices and to a large extent, to manage costs. Franchise contracts are generally in force for twenty years, after which they may be renewed, sold to new franchisees or the chain company may take over operations. The data for this study do not contain information about specific contract terms.

Franchisees with a single unit typically manage the restaurant themselves, generating strong incentives at the restaurant level. Franchisees who own multiple units cannot personally manage every restaurant that they own all of the time. At least some of their restaurants are operated, at least some of the time, by employees with fixed salaries and no residual claims. This generates weaker incentives than single-unit franchising. Multi-

unit franchise owners are, however, still more involved in the operation of individual restaurants than stockholders or higher level managers located at arms length, in terms of hierarchy and distance, for company-run units. This generates stronger incentives than company ownership. In this way, multi-unit franchising is intermediate between single-unit franchising and company ownership for the strength of incentives at the restaurant-level. I will examine the relative prevalence of multi-unit and single-unit franchising to help illuminate the nature of incentive problems in franchising.

In general, the assignment of residual claims to franchisees and the absence of residual claims for company managers generate different incentives for agents who perform similar tasks. Residual claims play a role in shaping incentives because employment and franchise contracts cannot fully account for all contingencies in restaurant operations. Even though they can lay out detailed procedures such as exact cooking times or paths of inspection for managers to take through restaurants on precise schedules, the actual actions taken may be imperfectly observed or unverifiable for the purposes of upholding a contract. Customer service and quality are vital to profits and especially difficult to specify fully. Furthermore, the value of franchisee-franchisor relationships may also restrict franchisors' ability to impose strict control over franchisees' actions.

Restaurant managers under both forms of ownership generate more valuable input as production activities become more complex. They may also have more flexibility in making decisions and allocating effort as production activities increase. Residual claims should therefore have greater value in motivating them to work hard and may also have a larger effect on the balance of incentives across diverse tasks. Franchisees have stronger

incentives to take actions that raise private returns, which is desirable for franchisors if these actions increase their overall profit but undesirable if particular franchisees' actions reduce profits at other restaurants or diminish the franchisor's ability to sell franchises in the future. In case studies of franchise systems, Bradach (1998) notes that franchisees have an incentive to modify their services to suit local tastes while franchisors value uniformity across all restaurants, a key tension in the franchisor-franchisee relationship. External effects also arise because franchisees have an incentive to free-ride on brand names. In a panel of franchisors, including some major restaurant chains, Lafontaine and Shaw (2005) show that franchisors with higher brand name value target higher rates of company-ownership, indicating that this is a way for them to avoid free-riding by franchisees.

## 2.3 Incentives and Residual Claims

Much of the early theoretical literature on franchising focuses on moral hazard problems and the role of residual claims in motivating effort from store managers. In simple agency models, managers' efforts are not observable and they have an incentive to shirk. Shirking reduces profits for the chain company. By giving managers a share in profits through a franchise contract, a chain company generates market incentives for its managers and reduces their incentive to shirk. However, this compels managers to bear unwanted risk or increases the franchisor's incentive to free-ride on their efforts. Caves and Murphy (1976) propose franchising as an efficient solution to the tradeoffs between centralized brand control and decentralized production. Rubin (1978) and Matthewson

and Winter (1985) show that profit-sharing between the franchisor and franchisees generates efficient effort from both participants for their contributions to production. If moral hazard for the manager becomes a greater problem relative to these drawbacks when more tasks are added to a manager's responsibilities, then more complex production should be correlated with stronger incentives at the restaurant. This means that restaurants that offer more services should be franchised, not company-owned, so that they are operated by franchisees with strong incentives to work hard.

More recent theories recognize that strong incentives motivate hard work but may also distort an agent's allocation of effort across different activities in a way that is not optimal for the chain. Holmstrom and Milgrom (1991, 1994) derive principal-agent models that explain why fixed wage payments may be superior to performance-based incentives and why transactions within a firm may be more efficient than market transactions when performance measures are incomplete and agents have complex responses to performance-based compensation. This idea is applicable to chain restaurants since restaurant production involves a number and variety of tasks and restaurant managers often have considerable flexibility in making decisions that affect day to day operations. Residual claims in the form of franchising provide aggregate rewards for a set of often disparate responsibilities and may not sufficiently motivate intangible contributions to service and quality, cooperation with other units or long term planning related to the value of a chain as a whole. They may skew franchisee-managers' efforts towards a subset of jobs at the expense of other vital functions or give them an incentive for inefficient activities such as manipulating accounts. If more complicated

responsibilities allow greater scope for distorted incentives, then it is efficient for a manager with more tasks to have weaker incentives. The lack of residual claims for company managers may generate balanced incentives at restaurants that provide more services.

A chain company's decision to own or franchise each restaurant can be modeled as a principal-agent problem. The chain company seeks to maximize restaurant value by motivating optimal effort from its manager. It has a choice of two instruments, an employee contract with no residual claims or a franchise contract that grants residual profits. Residual claims in franchising give the franchisee strong incentives to take actions that raise restaurant value but might distort his efforts towards those that benefit himself at the expense of overall efficiency. The lack of residual claims under company ownership lowers a manager's incentives to work hard but also reduces the efficiency loss from unbalanced incentives. The optimal choice of contract form will depend on the type and quantity of his responsibilities and the relative importance of strong and balanced incentives. Simple moral hazard models and multitasking models generate contrasting predictions for the relationship between responsibilities and incentives. This paper examines these broad predictions by comparing restaurant characteristics to the choice between ownership and franchising.

## 3 Data Sources and Variables

The main data for this paper are establishment-level data from the 1997 Census of Retail Trade (CRT) for the food services industry, which contain information about

ownership, type of business, revenues, payroll, employees, location and sales by merchandise lines for virtually all food services establishments in the United States. They also contain operational information such as the availability of seating and seating capacity for large employers. The Longitudinal Business Database (LBD) provides names, age and firm identifiers for each establishment in the CRT.

Nation's Restaurant News special report "*Top 100 Chain and Company Rankings 2003*" and "*Second 100 Chain and Company Rankings 2003*" (NRN Top 200) is a list of the largest restaurant chains, ranked by total sales and by total number of establishments nationwide. This source provides chain names for identifying chains in the CRT and an industry standard business concept for each chain. The CRT data contain observations for a majority of chains in NRN Top 200. These were identified by matching on chain and establishment names. A small number of other chains were separately identified in the CRT data. The remaining establishments in the CRT are classified as independent restaurants and provide information about market conditions such as the total number of restaurants in each county.

The dependent variable in the analysis in this paper is organizational form, whether a chain establishment is company-owned (CO) or franchised (FR). Each chain system consists of a franchisor, also called the chain company, and a set of company-owned and franchised restaurants. I identify company-owned restaurants using unique firm identifiers for franchisors. Other restaurants bearing the chain name are designated as franchises. Franchises can be further characterized as multi-unit franchises (MUF) or single-unit franchises (SUF), based on firm identifiers for franchisees. MUF consist of a

group of restaurants operated by a single franchisee. SUF are franchisees with only one outlet each.

The key independent variables in this study measure cross-chain and within-chain variation in the complexity of restaurant production, reflecting the quantity and variety of tasks supervised by restaurant managers. Restaurant production varies across chains because chains offer different types of food products and service formats that determine how much food preparation and customer service is produced at each restaurant. The production technology for each chain is broken down into two indicator variables. The food preparation indicator, ONSITE, is equal to 1 if a significant amount of cooking or other food preparation takes place at the restaurant and equal to 0 otherwise. The full service indicator, FULL, is equal to 1 for full service restaurants and equal to 0 for all other types of businesses. These variables divide chain restaurants into three groups, with additional tasks as each indicator goes from 0 to 1. Snack shops, defined as ice-cream shops, frozen yogurt shops and donut shops with baking off-premises, have no food production and no full service. Limited service restaurants, which include most fast food sandwich and pizza chains, donut shops with baking on-premises, bagel shops, coffee shops and bakeries carry out food production but do not provide table service. Full service restaurants perform both food production and full table service. ONSITE and FULL classify the full sample of chains.

Production activities also vary from establishment to establishment within a chain and across chains within a service category. Variables such as primary type of service, seating facilities and price range as a proxy for quality capture these differences. The

variable, TAKEOUT/DRIVETHRU, is set to 1 for restaurants whose primary food service type is takeout or drive-through service and 0 for other primary food service types. The variable, DELIVERY, is set to 1 for restaurants that focus on delivery services and 0 for other primary food service types. Limited service restaurants with TAKEOUT/DRIVETHRU equal to 0 and DELIVERY equal to 0 typically list dine-in service as their primary food service type. I shall consider dine-in restaurants to have more complex production activities than takeout, drive-through and delivery restaurants.

The availability of seats also captures the extent to which a restaurant serves dine-in customers rather than off-site consumers. The seating facilities indicator, SEATING, is set to 1 for limited restaurants that provide seating and 0 for those that do not. Notice that the variables, TAKEOUT/DRIVETHRU, DELIVERY and SEATING, are correlated in the sense that restaurants with no seating probably focus on takeout, drive-through or delivery but restaurants that are primarily takeout, drive-through and delivery may also offer seated service. I assume that restaurants with SEATING equal to 1 have more complicated production tasks, mostly due to more customer service, than those with SEATING set to 0.

In full service restaurants, price range generally reflects the amount of work that goes into food production and customer service. Average meal price varies little within chains but varies greatly across chains in full service. The indicator variable, PRICEO10, captures this variation. PRICEO10 is equal to 1 if average meal price is over $10 and equal to 0 if average meal price is $10 or under. This separates basic table service chains from higher quality chains that engage in more sophisticated food production and

customer service. Restaurant managers in restaurants with PRICEO10 equal to 1 should have more complex responsibilities than those in restaurants with PRICEO10 equal to 0. The information on primary service type, seating and price range exists for a subset of establishments and the Census provides sampling weights.

In addition, the 1994 County and City Data Book (CCDB) provides market demographics for each county that can account for variation in the demand for food service from restaurants. The market characteristics variables are total population, population density, percent growth in population from 1980-1990, percent black, percent Hispanic, percent foreign-born, percent between ages 25 to 65, percent households with income between $35-75k, percent households with income over $75k, percent one person households, percent married with children households, average travel time to work and crimes per pop. Total population and population density measure total market size. The growth rate of population reflects growth in the demand for restaurants. Population between ages 25 and 65 and households with higher incomes are those who are likely to be restaurant customers and measure the demand for restaurants. Composition by race, household characteristics and average travel time to work also reflect variation in the demand for restaurants. The number of crimes per head is correlated to a lower propensity to go out and lower demand for restaurants. Additional variables from the CRT and other sources that account for chain and market characteristics are described as they occur in the following discussion.

# 4 Empirical Analysis

The empirical tests in this paper compare the choice between company ownership and franchising to characteristics of restaurant production that vary across chains and within chains. Frequency counts of establishments by ownership form and production activities show how these variables are related in the raw data. Logit analysis draws out the relationship between residual claims and managerial responsibilities, accounting for variation in market conditions, chain and restaurant-specific attributes, sampling biases and variability across and within chains. I also examine the relative incidence of multi and single-unit franchising among franchised units as production activities vary.

## 4.1 Cross-Chain Variation in Production Technology

The variables ONSITE and FULL describe the basic production technology in each chain. Table 1 cross-categorizes establishments by chain-wide production technology and organizational form. The number of company-owned units relative to franchises increases by more than 24% as ONSITE changes from 0 to 1. It increases by 34% when FULL changes from 0 to 1. Asterisks denote data for a small number of firms or establishments that are suppressed to protect confidential firm-level or establishment-level information. Company ownership is more common among limited service restaurants than snack shops and more common in full service restaurants than limited service outlets. These facts indicate that company ownership is more common among restaurants with more complex production activities.

In order to account for variation across chains, I calculate the share of company-owned establishments for each chain and show the distribution of chains across share classes in each production type category. Figure 1 contains histograms for chains with 0-20%, 20-40%, 40-60%, 60-80% and 80-100% company-ownership shares for snack shops, limited service and full service, with the last two ownership-share classes joined together for snack shops to protect confidential firm data. There are relatively more chains with high company-ownership shares in full service than in limited service than in snack shops. The average snack chain owns 27% of its establishments, the average limited service chain owns 48% of its establishments and the average full service chain owns 71% of its establishments. These facts also indicate that company ownership shares are higher in chains with more complex production activities.

For more precise analysis, the chain company's decision can be modeled as a discrete choice problem between company ownership (CO) and franchising (FR). Suppose that a chain company chooses, for each of its restaurants, the organizational form that generates greatest profit, conditional on the types of activities performed at the restaurant, chain characteristics and local market conditions. For each restaurant in the chain, it can choose to sell franchise rights to an independent agent, granting this agent residual claims on restaurant profits and generating strong market-driven incentives for the agent. The franchisee usually manages the restaurant directly or keep a close eye on restaurant operations even if he is not physically present at all times. Alternatively, the chain company can run the restaurant itself, retaining residual claims to restaurant profits and hiring a manager to take care of restaurant-level operations. The employee-manager will

have few residual claims and weak market-driven incentives. The impact of ownership incentives on profits depends on the types of tasks that the manager performs, that is, on restaurant characteristics.

For each restaurant indexed by subscript *i*, the chain company chooses a unique organizational form *j*, where $j \in \{CO, FR\}$. Let restaurant *i*'s profit under organizational form *j* be denoted by $y_{ij}$ and let the chain company's decision to own rather than franchise be denoted by $Y_i$. Restaurant profits are described by the reduced form equation,

$$y_{ij} = \beta_j' x_i + \varepsilon_{ij} \quad \forall\ j \in \{CO, FR\}$$

where $x_i$ is a vector of observed restaurant characteristics, including types of restaurant activities, chain characteristics and local market conditions, and $\varepsilon_{ij}$ represents an unobserved, random component of profits. The chain company chooses company ownership if it generates greater profits than franchising. The probability that restaurant *i* is company-owned is given by

$$P_i \equiv \Pr(Y_i = 1) = \Pr(y_{iCO} \geq y_{iFR})$$

For the purposes of estimation, assume that $\{\varepsilon_{ij}\}$ are independent Weibull random variables to obtain a logit model of the chain company's discrete choice problem. Therefore, the probability that a restaurant with characteristics $x_i$ is company-owned is

$$P_i = \Pr(Y_i = 1) = \frac{e^{\beta' x_i}}{1 + e^{\beta' x_i}}$$

The parameters, $\beta = \beta_{CO} - \beta_{FR}$, reflect the relative value of company-ownership over franchising for various restaurant attributes, chain and market characteristics.

Table 2 contains the estimated coefficients from models with different combinations of right-hand side variables, with robust standard errors that allow for correlated variability within chains but not across chains. Model (1) is a basic specification with ONSITE, FULL and market characteristics such as total population, age, race, income and household characteristics included on the right-hand side to control for variation in demand for different types of food service. The estimated coefficients for ONSITE and FULL are both positive and significant at the 1% level. The predicted probability for company ownership in the median restaurant increases by 26% for ONSITE and 35% for FULL for a discrete change from 0 to 1 for each variable. This means that the probability of company ownership is higher for limited and full service outlets compared to snack shops and for full service relative to limited service restaurants.

Models (2) through (8) account for omitted variation in several restaurant, chain and market characteristics, while model (9) is a benchmark model containing a combination of these controls. Model (2) shows that higher revenues are positively correlated with company ownership but do not change the estimated effects of onsite food production and full service. Model (3) controls for location specific variation in costs and benefits of company ownership that depend on franchising laws, health codes and other state-level regulation by adding dummy variables for each state in the data. The estimated coefficients for ONSITE and FULL do not change in a significant way. Model (4) adds the variable, UFOC, an indicator for the 13 states that require chain companies to file

Uniform Franchise Offering Circulars (UFOC) describing contract terms to franchise in those states. This raises the cost of franchising and could reduce the propensity to franchise in these states. In fact, UFOC is estimated to reduce, not raise, the likelihood of company-ownership. Furthermore, the estimates for ONSITE and FULL do not change significantly. Omitted state-level variation in regulation does not account for the positive relationship between production technology and company ownership.

Model (4) also contains market level variables DE, NV and SHARE INCORPORATED to account for differences in market-level risk that may affect the decision to franchise. Incorporation reflects market risk because it measures the extent to which restaurant owners wish to limit their personal liability from restaurant losses. Restaurants are typically sole proprietorships, partnerships or corporations. Corporations differ from the other two forms mainly in having limited liability. It costs a small amount of time and money to file incorporation documents with state governments and it will be worthwhile to do so if expected losses are large or frequent. It is cheaper to incorporate in Delaware and Nevada, states represented by the indicator variables, DE and NV, respectively. Uncertainty in demand and costs may vary from market to market within a chain and may systematically vary across chains if they are not uniformly distributed across markets. Chains that require more food preparation or service could also be riskier because of the added tasks. Furthermore, it is possible that chain companies or franchisees are better at handling risk because of deep pockets, better local knowledge or other non-incentives factors. This would generate a spurious link between production technology and organizational form. The results from Model (4) demonstrate that this is

not the case. The coefficient estimates for ONSITE and FULL do not vary much with the inclusion of these variables. The incorporation rate is correlated with franchising, suggesting that franchises may be better at handling county-level market risk. Notice, however, that the number of crimes per pop is positively and significantly correlated with company ownership in all specifications, suggesting that company ownership is optimal in riskier counties. The indicator variables, DE and NV, are not statistically significant and are dropped for subsequent analysis.

Model (5) adds the total number of restaurants and total number of chain restaurants in each county to account for variation in competitive environments. Chains may allocate residual claims for strategic reasons. Residual claims in franchising may motivate franchisees to be tough competitors while low profit incentives in company-owned units may motivate store managers to be soft competitors, either of which may be desirable under different market conditions. The results show that strategic considerations do not account for the correlation between ONSITE and FULL and company ownership. The total number of restaurants has no significant explanatory power for organizational form. The total number of chain restaurants is positively correlated with company ownership, which may indicate that chains generate soft competition through weak incentives in markets with many other chain restaurants.

Models (6), (7) and (8) include measures of chain expansion and vintage. It is possible that credit constraints on the chain owner or other history-dependent factors cause chains to expand by franchising and that these non-incentives reasons for franchising vary systematically by type of food service. If this is the case, measures of

outlet expansion and vintage should account for spurious correlation between type of food service and organizational form. The results from these models show that the positive estimates for ONSITE and FULL remain of similar sign and magnitude with the inclusion of expansion and vintage effects. The interpretation about expansion and vintage in this section should be taken loosely since research about chain expansion, such as Lafontaine and Shaw (2005), shows that expansion by franchising is not monotonic over a chain's lifecycle and would not be captured in this model. The total number of establishments in a chain and overall age of a chain are correlated with franchising, suggesting that chains expand by franchising. The total of number of same-chain establishments in a county, chain age in a county and establishment age are positively correlated with company ownership, suggesting that chains start out by setting up company-owned units. The first result also suggests that company ownership may generate soft competition among same-chain stores.

The key outcome of this analysis is that the estimated correlation between company ownership and onsite food production and table service does not vary in a significant way across the eight models. The positive relationship between company ownership and production activities at chain restaurants persists after accounting for omitted variation such as regulatory differences, market risk, competitive considerations, expansion and vintage that may affect the costs and benefits of company ownership. Model (9) contains a combination of these variables and serves as a benchmark model. In this model, the predicted probability of company ownership for the median restaurant increases by 28%

when there is onsite food production and increases further by 26% when there is table service.

Taking the chain as an observation, the predicted share of company-owned establishments in a chain is also higher when the chain engages in onsite food production and table service, after accounting for market variation, chain expansion, vintage and establishment size. I generate predicted shares of company ownership for each chain by taking the average of establishment-level predicted probabilities from Model (9) across establishments in the chain. Figure 2 shows how these predicted shares vary for chains in each production technology category. Chains in the snack shop category fall mostly in the 0-20% range for company ownership, chains in limited service fall mostly within 20-60%, while chains in full service fall in the 60-100% range. The average snack chain owns an estimated 7% of its establishments, the average limited service chain owns 42% of its establishments and the average full service chain owns 77% of its establishments. These results show that the positive relationship between more complex production activities in chains and more company ownership is a robust feature of the restaurant industry.

## 4.2 Within-Chain and Within-Service Variation in Production Activities

The variables TAKEOUT/DRIVETHRU, DELIVER and SEATING describe variation in establishment-level activity across restaurants in a chain. The variable PRICEO10 captures variation in production activities across chains within full service. Table 3 cross categorizes establishments by these variables and organizational form. For

limited service restaurants, the proportion of company-owned non-takeout-drive-through outlets is 20% more than the proportion of company-owned takeout-drive-through establishments. The proportion of company-owned non-delivery outlets is 21% more than the proportion of company-owned delivery outlets. There is more company ownership for dine-in restaurants than for takeout, drive-through and delivery restaurants. Taken together with the argument that dine-in service generates more complex production tasks than takeout, drive-through and delivery service alone, these facts show that more complex production modes are correlated with more company ownership.

The statistics in Table 3 also show that the proportion of company-owned outlets with seats is 23% greater than the proportion of company-owned outlets with no seats for limited service restaurants. Company ownership is more common among restaurants with dine-in service than for restaurants that do not accommodate dine-in customers.

The last two rows of Table 3 show that for full service restaurants, the number of company-owned units relative to franchised units is 30% greater in those with prices over $10 than in those with prices under $10. There is more company ownership among more expensive full service restaurants. If price is positively correlated with the quality of meals and service, this fact indicates that company ownership is more common for restaurants with more complex production tasks. As a whole, the statistics in Table 3 show that company ownership and more complex production activities are positively correlated for activities that vary within chains and service categories, matching the findings for activities that vary across chains.

To account for restaurant and market-specific variation, I estimate logit models similar to Model (9) in Table 2 for these production characteristics, with weighted observations to take account of sampling bias. Table 4 contains the estimated coefficients and robust standard errors. The indicator variables, TAKEOUT/DRIVETHRU, DELIVERY, SEATING and PRICEO10, measure the effect of production activities on the probability of company ownership. In addition to market demographic information that controls for characteristics of demand, right-hand side variables that account for establishment characteristics are total revenues, age, number of own-chain establishments in the county and chain fixed effects where appropriate, and those that control for market conditions are UFOC, share incorporated, total establishments and total chain establishment.

The results for Model (1) show that the estimated coefficient for TAKEOUT/DRIVETHRU is negative but not significant, while that for DELIVERY is negative and significant at the 1% level. The predicted probability of company ownership decreases by 13% for takeout-drive-through outlets and by 22% for delivery outlets. Limited service chains are less likely to own establishments with no dine-in service than those with dine-in service. After controlling for differences in market characteristics, chain identity, age and size, dine-in restaurants within a limited service chain system are more likely to be company-owned than takeout-drive-though and delivery outlets.

The results for Model (2) show that the estimated coefficient for SEATING is positive and significant at the 1% level. The predicted probability of company ownership increases by 18% for restaurants that have seats over those with no seats. Within a

limited service chain, restaurants with dine-in service are more likely to be company-owned than restaurants that do not provide seating, reiterating the findings from Model (1).

For the sample of full service restaurants in Model (3), the estimated coefficient for PRICEO10 is positive and significant at the 1% level. The predicted probability for company ownership increases by 24% for full service restaurants with average price greater than $10 relative to those with average price under $10. Since there is little price variation within chains, this is largely a cross-chain effect. Restaurants in full service chains with higher prices are more likely to be company-owned than those with lower prices. If prices reflect the range and quality of production activities, then company ownership and complexity of production tasks are positively linked when comparing chains within full service, as well as chains across different service categories.

The results in this section suggest that company ownership is systematically correlated with production complexity within limited service chains and within full service, as well as across chains with different production technology. This finding is not consistent with the view that strong incentives in franchising simply motivate managers to work hard. It is consistent with the view that the balance of incentives matters; company ownership reduces the scope for undesirable actions by limiting residual claims when there are more restaurant-level activities.

## 4.3 Multi-Unit and Single-Unit Franchising

The relationship between production activities and multiple-unit franchising sheds additional light on the nature of incentive problems in decentralized production. Establishment counts in Table 5 show that multi-unit franchising is the predominant form of franchising among chain restaurants, accounting for over 50% of franchised establishments in nearly every production-type category. The high proportion of multi-unit owners provides further evidence that the main benefit of franchising is not simply to solve monitoring problems at the restaurant level. The principal, in this case, is the multi-unit franchisee who must still motivate managers to work hard, even if moral hazard is mitigated by closer monitoring or family ties.

A candidate explanation for why company ownership increases with production activities is that residual claims skew franchisees' efforts towards their private gains at the expense of the chain company's profits. Multi-unit franchisees should have less incentive than single-unit franchisees to free-ride but more incentive to hold up the franchisor for private gains. Multiple-unit owners internalize spillovers among several restaurants, lessening their incentive to free-ride on the brand name or otherwise reduce brand value. However, they have more bargaining power, which increases their incentives for hold-up activities. They also have greater market power, exacerbating double marginalization problems. I compare the relative incidence of multi-unit and single-unit franchising with different production activities to determine if either incentive problems due to externalities or hold-up issues dominate as disadvantages of franchising.

Table 5 shows raw establishment counts for multi-unit and single-unit franchises in each production-type category. With the exception of onsite food production, additional production activities are correlated with lower proportions of multi-unit franchising. Snack shops have the lowest proportion of multi-unit franchise establishments at 29%. Limited service and full service restaurants both have 70% multi-unit franchising among franchisees. The addition of onsite food production goes together with an increase in multi-unit franchising. The addition of table service, on the other hand, is not associated with changes in the form of franchising. Among limited service restaurants, dine-in restaurants have slightly lower proportions of multi-unit franchising than takeout-drive-through, delivery and seatless restaurants. For full service chains, restaurants with prices over $10 have a lower proportion of multi-unit franchising than restaurants with prices under $10.

These patterns remain after controlling for restaurant and market characteristics in logit analysis. Table 6 contains estimated coefficients and robust standard errors for four logit models run on the subsample of franchised establishments. The dependent variable in each specification is MUF, set to 1 for multi-unit franchised establishments and set to 0 for single-unit franchised outlets. Model (1) results show that the estimated parameter for ONSITE is positive and significant at the 1% level. The estimated coefficient for FULL is negative and not significantly different from zero. The predicted probability of multi-unit franchising increases by 33% for a discrete change from 0 to 1 in ONSITE.

For limited service franchises, Models (2) and (3) show that production modes with more onsite customer service are associated with less multi-unit franchising among

franchises. Exclusion to TAKEOUT/DRIVETHRU and DELIVERY services is statistically correlated with multi-unit franchising at the 10% and 1% levels, respectively. SEATING has no statistically significant effect on the form of franchising.

For full service franchises, more expensive restaurants are less likely to be part of multi-unit franchises. Model (4) generates a negative and significant estimated coefficient for PRICEO10 in full service franchises. The predicted probability of multi-unit franchising decreases by 40% for restaurants with average price over $10.

These results indicate that more complex production activities are associated with less multi-unit franchising among franchisees. This goes together with more company ownership in the chain system as a whole, suggesting that incentive problems in franchising are not mitigated by multi-unit ownership for franchisees.

# 5 Conclusion

This paper examines the question of whether production more likely takes place within the firm or through market transactions as production activities become more complex. Restaurant chains provide an exceptional environment for studying this question because they generally employ two forms of organization, company ownership and franchising, that differ in the strength of market incentives. Furthermore, production activities vary widely across different chains while remaining largely uniform within chains. With access to Economic Census data with establishment-level records for entire

chain systems, I am able to directly compare the choice of incentives for managers to the types of tasks that they perform.

The main finding is that company ownership, rather than franchising, is more common for restaurants with more complex production activities. Company ownership is most common in full service chains, followed by limited service chains, and least common for snack chains. Looking across different types of restaurant chains, onsite food production raises the likelihood of company ownership by 28% relative to offsite food production while table service raises the likelihood of company ownership by 26% relative to counter service. Among limited service restaurants, company ownership is more common in outlets with dine-in services than in those with no dine-in services. Restaurants that serve onsite customers are 13% and 22% more likely to be company-owned than restaurants that serve only takeout-drive-through and delivery customers, respectively. Restaurants that provide seating are 18% more likely to be company-owned than those with no seating. I argue that dine-in customers generate more complex managerial tasks by requiring more input into customer service. For full service chains, company ownership is more common among chains with higher average prices. Restaurants with average prices over $10 are 24% more likely to be company-owned than those with prices below $10. Since average prices are correlated with quality, this implies that company ownership is more common in restaurants where food preparation and customer service are more intricate.

Furthermore, the increased incidence of company ownership in chains with more complex production activities goes together with a reduction in the use of multi-unit

franchising. This suggests that incentive problems in franchising are not easily solved by internalizing the spillovers among small groups of restaurants.

These facts show that restaurant managers with more responsibilities tend to have fewer residual claims to the profits from their efforts. Weak, rather than strong, incentives are associated with more complicated production processes, indicating that the choice between in-house production and franchising in restaurant chains does not hinge on simple shirking, which should be more tempting for managers as their responsibilities grow. Modern theory such as multi-task models by Holmstrom and Milgrom (1991, 1994) and Hart, Shleifer and Vishny (1997) suggest that company ownership in restaurant chains serves to balance restaurant managers' incentives across tasks as production activities grow more involved and managers find more scope to allocate their efforts towards private gains. By severing the link between private returns and effort, company ownership may suppress incentives for activities such as free riding and modifications solely for local tastes that are pervasive problems in franchise systems. This interpretation of my findings is consistent with growing evidence in empirical research by Shepard (1993), Slade (1996), Hubbard (2000), Baker and Hubbard (2003, 2004), and Lafontaine and Shaw (2005) that ownership incentives play a complex role in coordinating production and circumventing problems due to market interactions in a variety of industries and contractual situations.

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# Appendix Tables

Table 1. Establishment counts by production technology and organizational form

| Production technology | No. of establishments<br>Percentage of total establishments<br>CO | FR | Total establishments | No. of chains |
|---|---|---|---|---|
| ONSITE=0 FULL=0<br>*(Snack shops)* | *<br><10% | 4681<br>>90% | * | * |
| ONSITE=1 FULL=0<br>*(Limited service)* | 24201<br>34% | 47048<br>66% | 71249<br>100% | 72 |
| ONSITE=1 FULL=1<br>*(Full service)* | 12103<br>68% | 5798<br>32% | 17901<br>100% | 75 |
| Total establishments | * | 57527 | >90000 | <200 |

Figure 1. Frequency distribution of chains by percentage of CO.

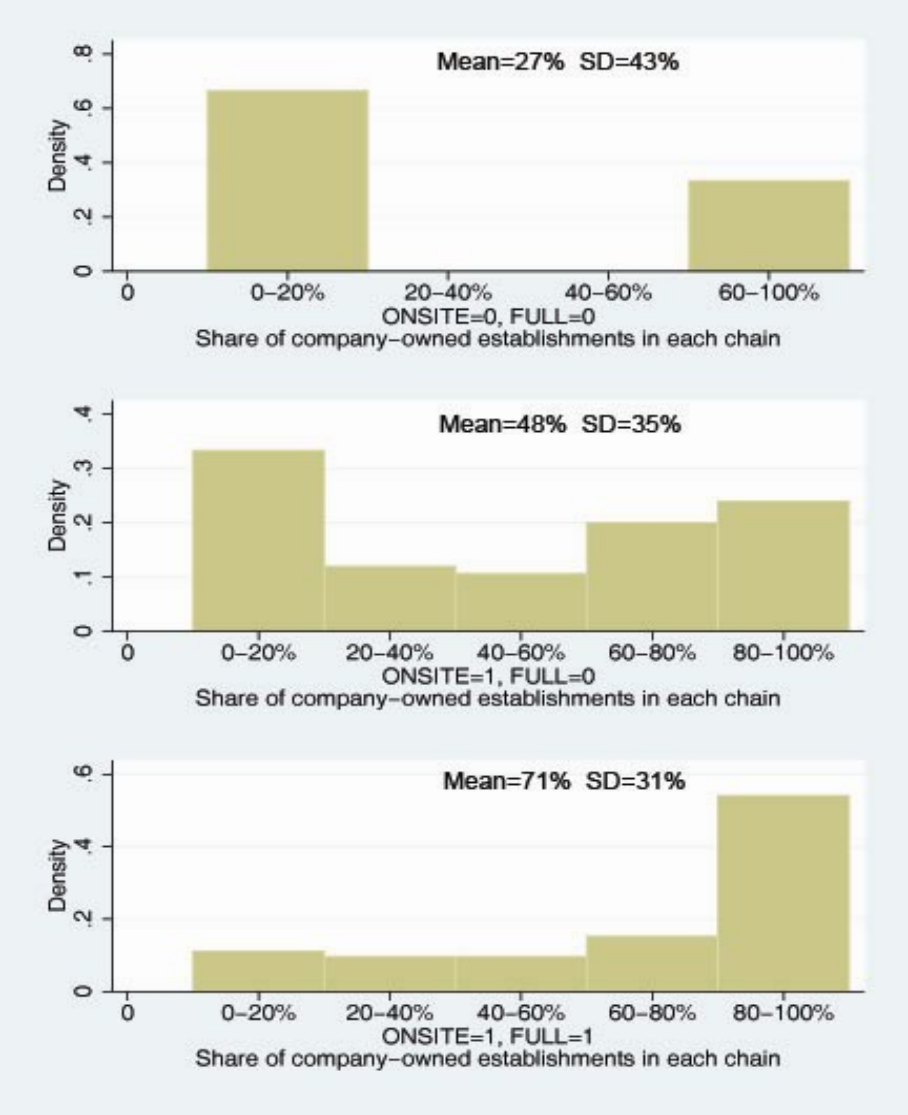


Table 2. Cross-chain variation in production technology and company ownership.
Logit parameter estimates and incremental effects estimates calculated at sample medians.

| | (1) | (2) | (3) | (4) | (5) | (6) | (7) | (8) | (9) |
|---|---|---|---|---|---|---|---|---|---|
| *Dependent variable* | P(CO=1) | P(CO=1) | P(CO=1) | P(CO=1) | P(CO=1) | P(CO=1) | P(CO=1) | P(CO=1) | P(CO=1) |
| Obs= | >90000 | >90000 | >90000 | >90000 | >90000 | >90000 | >90000 | >90000 | >90000 |
| Log likelihood= | -54433 | -54071 | -53293 | -53908 | -53744 | -52757 | -52579 | -52187 | -52242 |
| Pseudo R-sq= | 0.13 | 0.14 | 0.15 | 0.14 | 0.15 | 0.16 | 0.16 | 0.17 | 0.17 |
| ONSITE | 2.2310 | 2.1041 | 2.1340 | 2.1409 | 2.1140 | 2.3993 | 2.0424 | 1.9989 | 2.3949 |
| | 1.1095 | 1.1092 | 1.1062 | 1.1133 | 1.1013 | 1.0913 | 1.1037 | 1.0754 | 1.0596 |
| FULL | 1.4616 | 1.3008 | 1.3141 | 1.2953 | 1.3028 | 1.1528 | 1.3363 | 1.2473 | 1.0920 |
| | 0.3083 | 0.2942 | 0.2955 | 0.2945 | 0.2937 | 0.3065 | 0.3017 | 0.2942 | 0.2890 |
| Revenues ($k) | | 0.0031 | 0.0031 | 0.0031 | 0.0031 | 0.0030 | 0.0030 | 0.0029 | 0.0029 |
| | | 0.0025 | 0.0025 | 0.0025 | 0.0025 | 0.0023 | 0.0024 | 0.0024 | 0.0022 |
| State fixed effects | | | yes | | | | | | |
| UFOC | | | | -0.3215 | -0.2734 | -0.2743 | -0.2710 | -0.2716 | -0.2805 |
| | | | | 0.1064 | 0.1062 | 0.1090 | 0.1064 | 0.1074 | 0.1081 |
| DE | | | | yes | | | | | |
| NV | | | | yes | | | | | |
| Share incorporated | | | | -0.1498 | -0.2911 | -0.5958 | -0.4992 | -0.5479 | -0.7199 |
| | | | | 0.5248 | 0.5106 | 0.4907 | 0.5226 | 0.5133 | 0.4985 |
| Total establishments | | | | | -0.1525 | -0.1932 | -0.0928 | -0.1005 | -0.1607 |
| | | | | | 0.1525 | 0.1358 | 0.1501 | 0.1637 | 0.1377 |
| Total chain establishments | | | | | 1.4328 | 1.2525 | 1.1557 | 1.3235 | 1.1778 |
| | | | | | 0.3324 | 0.3311 | 0.3062 | 0.3402 | 0.3269 |
| Own chain total estabs | | | | | | -0.1243 | | | -0.1269 |
| | | | | | | 0.0750 | | | 0.0743 |
| Own chain estabs in county | | | | | | 0.0045 | | | 0.0040 |
| | | | | | | 0.0025 | | | 0.0025 |
| Own chain age | | | | | | | -0.1700 | -0.1580 | |
| | | | | | | | 0.0531 | 0.0465 | |
| Own chain age in county | | | | | | | 0.0338 | | |
| | | | | | | | 0.0185 | | |
| Establishment age | | | | | | | | 0.0450 | 0.0347 |
| | | | | | | | | 0.0140 | 0.0117 |
| *Market characteristics estimates, next page* | | | | | | | | | |
| Predicted P(CO=1) calculated at medians | 0.3088 | 0.3050 | 0.4044 | 0.3307 | 0.3161 | 0.3346 | 0.3046 | 0.2828 | 0.3257 |
| ΔP(CO=1) from discrete change in | | | | | | | | | |
| ONSITE | 0.2630 | 0.2542 | 0.3300 | 0.2758 | 0.2632 | 0.2909 | 0.2509 | 0.2321 | 0.2835 |
| FULL | 0.3495 | 0.3121 | 0.3121 | 0.3127 | 0.3136 | 0.2797 | 0.3204 | 0.2957 | 0.2644 |

Table 2 continued.

| | (1) | (2) | (3) | (4) | (5) | (6) | (7) | (8) | (9) |
|---|---|---|---|---|---|---|---|---|---|
| *Dependent variable* | P(CO=1) | P(CO=1) | P(CO=1) | P(CO=1) | P(CO=1) | P(CO=1) | P(CO=1) | P(CO=1) | P(CO=1) |
| Obs= | >90000 | >90000 | >90000 | >90000 | >90000 | >90000 | >90000 | >90000 | >90000 |
| Log likelihood= | -54433 | -54071 | -53293 | -53908 | -53744 | -52757 | -52579 | -52187 | -52242 |
| Pseudo R-sq= | 0.13 | 0.14 | 0.15 | 0.14 | 0.15 | 0.16 | 0.16 | 0.17 | 0.17 |
| Total population | 0.0001 | 0.0001 | 0.0001 | 0.0001 | -0.0001 | -0.0001 | -0.0002 | -0.0002 | -0.0001 |
| | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0001 | 0.0001 | 0.0002 | 0.0002 | 0.0001 |
| Population density | 0.0075 | 0.0036 | 0.0066 | 0.0088 | 0.0279 | 0.0282 | 0.0233 | 0.0275 | 0.0285 |
| | 0.0124 | 0.0115 | 0.0106 | 0.0121 | 0.0124 | 0.0120 | 0.0130 | 0.0125 | 0.0122 |
| Percent change in population | 0.0017 | 0.0017 | 0.0001 | 0.0008 | -0.0006 | -0.0011 | 0.0000 | 0.0003 | -0.0002 |
| | 0.0015 | 0.0015 | 0.0012 | 0.0015 | 0.0014 | 0.0015 | 0.0014 | 0.0014 | 0.0014 |
| Average wage | 0.0598 | 0.0503 | 0.0499 | 0.0502 | 0.0503 | 0.0552 | 0.0493 | 0.0439 | 0.0468 |
| | 0.0294 | 0.0325 | 0.0323 | 0.0324 | 0.0325 | 0.0350 | 0.0319 | 0.0339 | 0.0356 |
| Percent black | 0.0030 | 0.0031 | 0.0057 | 0.0020 | 0.0023 | 0.0016 | 0.0033 | 0.0027 | 0.0015 |
| | 0.0032 | 0.0033 | 0.0036 | 0.0033 | 0.0032 | 0.0033 | 0.0033 | 0.0033 | 0.0033 |
| Percent hispanic | 0.0000 | 0.0001 | 0.0059 | -0.0011 | -0.0046 | -0.0054 | -0.0062 | -0.0058 | -0.0060 |
| | 0.0058 | 0.0058 | 0.0057 | 0.0057 | 0.0059 | 0.0053 | 0.0058 | 0.0058 | 0.0052 |
| Percent foreign | -0.0008 | -0.0010 | -0.0042 | 0.0041 | 0.0115 | 0.0124 | 0.0137 | 0.0133 | 0.0129 |
| | 0.0086 | 0.0087 | 0.0081 | 0.0086 | 0.0087 | 0.0085 | 0.0092 | 0.0089 | 0.0086 |
| Percent age 24-54 | -0.0084 | -0.0091 | 0.0143 | -0.0139 | -0.0269 | -0.0285 | -0.0264 | -0.0265 | -0.0274 |
| | 0.0156 | 0.0158 | 0.0104 | 0.0155 | 0.0154 | 0.0149 | 0.0157 | 0.0160 | 0.0152 |
| Percent income $35-75k | 0.0366 | 0.0353 | 0.0434 | 0.0455 | 0.0460 | 0.0474 | 0.0381 | 0.0432 | 0.0477 |
| | 0.0133 | 0.0133 | 0.0071 | 0.0121 | 0.0115 | 0.0106 | 0.0121 | 0.0120 | 0.0103 |
| Percent income over $75k | 0.0169 | 0.0163 | 0.0200 | 0.0139 | 0.0117 | 0.0088 | 0.0074 | 0.0093 | 0.0093 |
| | 0.0073 | 0.0074 | 0.0070 | 0.0072 | 0.0069 | 0.0066 | 0.0074 | 0.0071 | 0.0068 |
| Crimes per pop | 0.0001 | 0.0001 | 0.0001 | 0.0001 | 0.0001 | 0.0001 | 0.0001 | 0.0001 | 0.0001 |
| | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 |
| Average travel time to work | 0.0366 | 0.0374 | 0.0214 | 0.0350 | 0.0336 | 0.0341 | 0.0348 | 0.0367 | 0.0362 |
| | 0.0104 | 0.0108 | 0.0091 | 0.0097 | 0.0100 | 0.0101 | 0.0103 | 0.0104 | 0.0105 |
| Percent one person households | 0.0207 | 0.0246 | 0.0075 | 0.0183 | 0.0073 | 0.0057 | 0.0051 | 0.0044 | 0.0063 |
| | 0.0155 | 0.0155 | 0.0082 | 0.0156 | 0.0156 | 0.0160 | 0.0161 | 0.0161 | 0.0161 |
| Percent married with children hou | -0.0087 | -0.0059 | -0.0193 | -0.0105 | -0.0123 | -0.0125 | -0.0059 | -0.0098 | -0.0125 |
| | 0.0134 | 0.0138 | 0.0074 | 0.0136 | 0.0136 | 0.0133 | 0.0145 | 0.0141 | 0.0135 |
| Constant | -6.1659 | -6.2986 | -5.9550 | -5.9584 | -5.0204 | -4.6792 | -1.5262 | -1.7695 | -4.9894 |
| | 1.1509 | 1.1339 | 1.3843 | 1.1567 | 1.1665 | 1.2054 | 1.7657 | 1.6752 | 1.1716 |

Figure 2. Frequency distribution of chains by average predicted probability of CO.

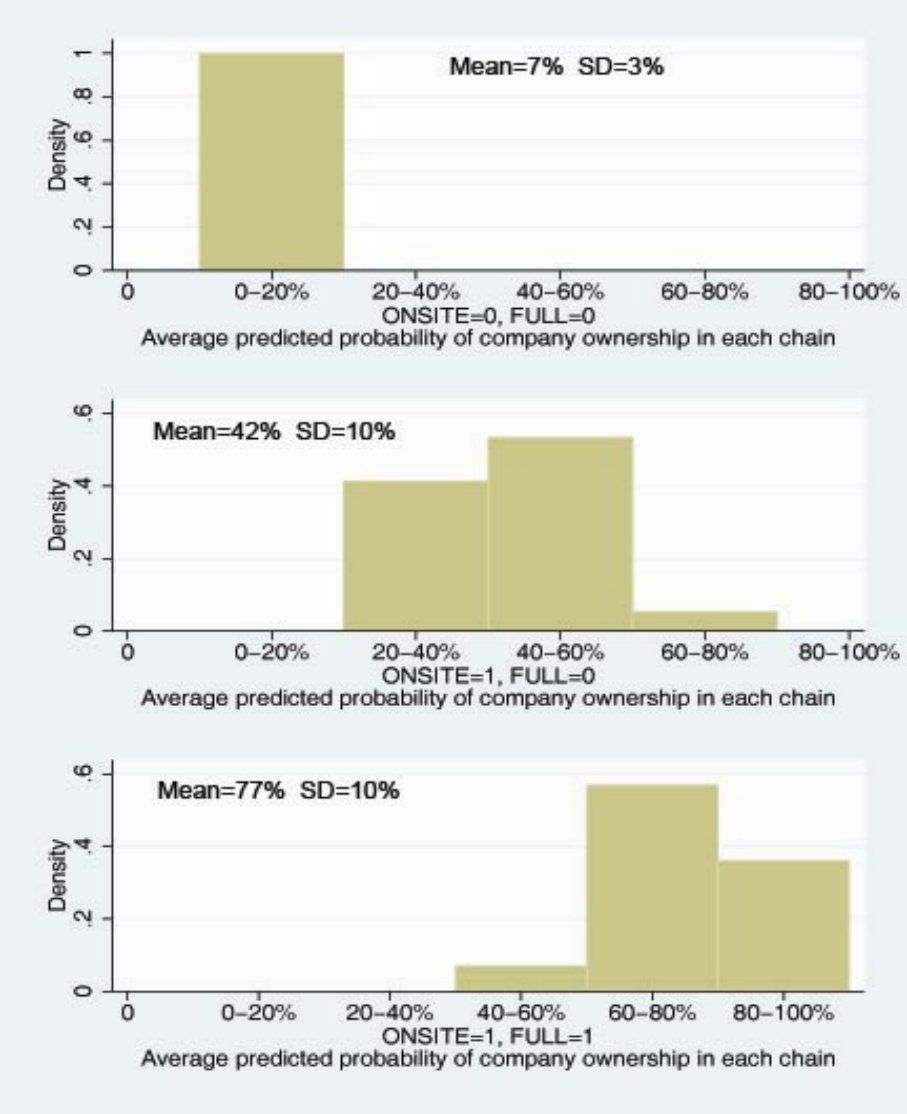


Table 3. Establishment counts by production mode and organizational form.

| | | No. of establishments<br>Percentage of total establishments | | Total establishments |
|---|---|---|---|---|
| | | CO | FR | |
| *Limited Service* | TAKEOUT/DRIVETHRU=1 | 2719<br>21% | 10001<br>79% | 12720<br>100% |
| | TAKEOUT/DRIVETHRU=0 | 17386<br>41% | 25023<br>59% | 42409<br>100% |
| *Limited Service* | DELIVERY=1 | 539<br>17% | 2650<br>83% | 3189<br>100% |
| | DELIVERY=0 | 19566<br>38% | 32374<br>62% | 51940<br>100% |
| *Limited Service* | SEATING=0 | 557<br>10% | 5224<br>90% | 5781<br>100% |
| | SEATING=1 | 15909<br>33% | 31928<br>67% | 47837<br>100% |
| *Full Service* | PRICEO10=0 | 6994<br>61% | 4409<br>39% | 11403<br>100% |
| | PRICEO10=1 | 1973<br>91% | 185<br>9% | 2158<br>100% |

Table 4. Within-chain and within-service variation in production mode and company ownership. Logit parameter estimates and incremental effects estimates calculated at sample medians.

| | (1) | (2) | (3) |
|---|---|---|---|
| *Dependent variable* | P(CO=1) | P(CO=1) | P(CO=1) |
| Obs= | 55129 | 43156 | 13561 |
| Log likelihood= | -24858 | -17802 | -7038 |
| Pseudo R-sq= | 0.30 | 0.36 | 0.21 |
| TAKEOUT/DRIVETHRU | -0.9203 | | |
| | 0.8093 | | |
| DELIVERY | -2.6171 | | |
| | 1.0072 | | |
| SEATING | | 2.6826 | |
| | | 1.1582 | |
| PRICEO10 | | | 1.2433 |
| | | | 0.6960 |
| Revenues ($k) | 0.0006 | -0.0002 | 0.0090 |
| | 0.0018 | 0.0014 | 0.0028 |
| Establishment age | 0.0447 | 0.0422 | 0.0484 |
| | 0.0085 | 0.0098 | 0.0270 |
| Own chain total estabs in county | 0.0038 | 0.0030 | 0.0080 |
| | 0.0024 | 0.0028 | 0.0182 |
| Chain fixed effects | yes | yes | no |
| UFOC | -0.4996 | -0.3955 | -0.2374 |
| | 0.2069 | 0.2212 | 0.1894 |
| Share incorporated | -2.6690 | -1.7817 | 1.5626 |
| | 0.6341 | 0.7470 | 0.8362 |
| Total establishments | -0.2433 | -0.3575 | 0.2469 |
| | 0.2014 | 0.1882 | 0.3875 |
| Total chain establishments | 1.6504 | 1.9611 | 0.0202 |
| | 0.4388 | 0.4302 | 0.9619 |
| *Market characteristics estimates omitted* | | | |
| Predicted P(CO=1) | 0.2476 | 0.1925 | 0.5971 |
| ΔP(CO=1) from discrete change in | | | |
| TAKEOUT/DRIVETHRU | -0.1317 | | |
| DELIVERY | -0.2241 | | |
| SEATING | | 0.1764 | |
| PRICEO10 | | | 0.2400 |

Table 5. Establishment counts by production activities and franchise form.

| | | No. of establishments<br>Percentage of franchised establishments<br>MUF | <br><br>SUF | Total franchised establishments |
|---|---|---|---|---|
| *All restaurants* | ONSITE=0, FULL=0<br>*(Snack shops)* | 1351<br>29% | 3330<br>71% | 4681<br>100% |
| | ONSITE=1, FULL=0<br>*(Limited service)* | 33514<br>71% | 13534<br>29% | 47048<br>100% |
| | ONSITE=1, FULL=1<br>*(Full service)* | 4030<br>70% | 1768<br>30% | 5798<br>100% |
| *Limited Service* | TAKEOUT/DRIVETHRU=1 | 9025<br>84% | 1778<br>16% | 10803<br>100% |
| | TAKEOUT/DRIVETHRU=0 | 24971<br>78% | 7163<br>22% | 32134<br>100% |
| *Limited Service* | DELIVERY=1 | 2385<br>89% | 287<br>11% | 2672<br>100% |
| | DELIVERY=0 | 31611<br>79% | 8654<br>21% | 40265<br>100% |
| *Limited Service* | SEATING=0 | 4002<br>69% | 1222<br>21% | 5224<br>100% |
| | SEATING=1 | 25018<br>52% | 6910<br>14% | 31928<br>100% |
| *Full Service* | PRICEO10=0 | 3508<br>80% | 901<br>20% | 4409<br>100% |
| | PRICEO10=1 | 76<br>41% | 109<br>59% | 185<br>100% |

Table 6. Cross-chain and within-chain variation in production activities and franchise form.
Logit parameter estimates and incremental effects estimates calculated at sample medians.

| | (1) | (2) | (3) | (4) |
|---|---|---|---|---|
| *Dependent variable* | P(MUF=1) | P(MUF=1) | P(MUF=1) | P(MUF=1) |
| Obs= | 52527 | 42937 | 36052 | 4594 |
| Log likelihood= | -32516 | -17862 | -15098 | -2524 |
| Pseudo R-sq= | 0.10 | 0.33 | 0.33 | 0.10 |
| ONSITE | 1.3607 | | | |
| | 0.3078 | | | |
| FULL | -0.1710 | | | |
| | 0.3955 | | | |
| TAKEOUT/DRIVETHRU | | 0.2617 | | |
| | | 0.1696 | | |
| DELIVERY | | 0.8170 | | |
| | | 0.3638 | | |
| SEATING | | | 0.0961 | |
| | | | 0.3777 | |
| PRICEO10 | | | | -1.7383 |
| | | | | 0.6910 |
| Revenues ($k) | 0.0046 | -0.0032 | -0.0031 | 0.0018 |
| | 0.0034 | 0.0010 | 0.0009 | 0.0025 |
| Establishment age | 0.0046 | 0.0031 | 0.0048 | -0.0122 |
| | 0.0122 | 0.0107 | 0.0107 | 0.0230 |
| Own chain total estabs in county | -0.0012 | 0.0049 | 0.0035 | 0.0387 |
| | 0.0033 | 0.0036 | 0.0032 | 0.0316 |
| Chain fixed effects | no | yes | yes | no |
| UFOC | 0.0455 | -0.0319 | -0.0424 | -0.5787 |
| | 0.0945 | 0.0747 | 0.0878 | 0.3682 |
| Share incorporated | 2.6565 | 2.4664 | 2.8926 | 1.7704 |
| | 0.3516 | 0.4720 | 0.4472 | 1.4686 |
| Total establishments | 0.0929 | 0.0919 | 0.0674 | 1.2250 |
| | 0.1671 | 0.2826 | 0.3003 | 0.4823 |
| Total chain establishments | 0.0386 | 1.4111 | 1.6049 | -1.7019 |
| | 0.3601 | 0.7572 | 0.8013 | 0.9883 |
| *Market characteristics estimates omitted* | | | | |
| Predicted P(MUF=1) | 0.7014 | 0.8535 | 0.8701 | 0.7563 |
| ΔP(MUF=1) from discrete change in | | | | |
| ONSITE | 0.3254 | | | |
| FULL | -0.0370 | | | |
| TAKEOUT/DRIVETHRU | | 0.0298 | | |
| DELIVERY | | 0.0760 | | |
| SEATING | | | 0.0113 | |
| PRICEO10 | | | | -0.4033 |